EduMadras

UI/UX Design Specification

For Stitch MCP / Design Handoff

India's Premier College Discovery Platform

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1. Brand Identity

EduMadras is a trusted college discovery platform for Indian students aged 17-24. The brand communicates reliability, clarity, and ambition. Every design decision must reinforce trust — students are making life-defining decisions on this platform.

1.1 Color Palette

The palette is derived directly from the EduMadras logo — navy blue, teal, and slate gray. These colors convey academic authority and modern professionalism.

Swatch	Color Name	Usage
#1B3A5C	**Navy Primary**	Headers, navbar, primary CTAs, hero backgrounds, active states
#1a7a6e	**Teal Accent**	Section accents, success states, teal hover effects, links
#2A5298	**Blue Mid**	Secondary buttons, info badges, hyperlinks
#F97316	**Orange Spark**	Urgency tags, offer banners, deadline alerts only
#E6F1FB	**Engineering Blue**	Engineering stream badge backgrounds
#E1F5EE	**Medical Green**	Medical stream badge backgrounds
#EEEDFE	**Management Purple**	Management stream badge backgrounds
#FAECE7	**Law Coral**	Law stream badge backgrounds
#FAEEDA	**Design Amber**	Design stream badge backgrounds
#F9FAFB	**Surface Gray**	Page background, card surfaces
#E5E7EB	**Border Light**	Card borders, dividers, input borders
#111827	**Text Primary**	All headings, primary body text
#6B7280	**Text Secondary**	Supporting text, metadata, labels
#9CA3AF	**Text Tertiary**	Placeholders, timestamps, hints
#FFFFFF	**White**	Card backgrounds, modal backgrounds, nav background

1.2 Typography System

EduMadras uses Inter as the primary typeface — a clean, highly legible sans-serif that renders well at all sizes on mobile and desktop. System font stack as fallback.

Element	Size	Weight	Usage
Hero Title	48px / 3rem	700 Bold	Homepage hero headline only

Page Title (H1)	`36px / 2.25rem`	700 Bold	College name, page headings
Section Title (H2)	`28px / 1.75rem`	600 SemiBold	Section headers on pages
Card Title (H3)	`20px / 1.25rem`	600 SemiBold	College card names, widget titles
Body Large	`16px / 1rem`	400 Regular	Primary body text, descriptions
Body Small	`14px / 0.875rem`	400 Regular	Card metadata, supporting info
Label	`13px / 0.8125rem`	500 Medium	Form labels, filter chips, tabs
Caption	`12px / 0.75rem`	400 Regular	Timestamps, image captions, footnotes
Badge / Tag	`11px / 0.6875rem`	600 SemiBold	Stream badges, status pills, tags
Navigation Link	`14px / 0.875rem`	500 Medium	Navbar links
Button Text	`14px / 0.875rem`	600 SemiBold	All button labels
Mono / Code	`13px / 0.8125rem`	400 Regular	Fee amounts when tabular, slugs

1.3 Spacing & Grid System

EduMadras uses an 8px base unit system. All spacing values are multiples of 8. The layout uses a 12-column grid at 1280px max-width with 24px gutters.

Token	Value — Usage
space-1	4px — Icon gaps, inline element spacing
space-2	8px — Component internal padding
space-3	12px — Card internal gaps
space-4	16px — Card padding, form field gaps
space-5	20px — Section subitem spacing
space-6	24px — Card margin, grid gutter
space-8	32px — Section spacing
space-10	40px — Large section padding
space-12	48px — Hero padding
space-16	64px — Page section separation
radius-sm	6px — Badges, pills, small buttons

radius-md	10px — Input fields, small cards
radius-lg	14px — College cards, modals
radius-xl	20px — Hero sections, large containers
radius-full	9999px — Stream filter pills, avatar circles
shadow-card	0 1px 3px rgba(0,0,0,0.08) — College cards default
shadow-hover	0 8px 24px rgba(0,0,0,0.12) — College cards on hover
shadow-modal	0 20px 60px rgba(0,0,0,0.20) — Modals, dropdowns

1.4 Logo Usage

- Minimum size: 120px wide on desktop, 80px on mobile
- Clear space: Equal to the height of the "E" in EduMadras on all sides
- Dark backgrounds: Use full-color logo (navy + teal)
- White navbar: Use full-color logo
- NEVER stretch, rotate, recolor, or add effects to the logo
- Favicon: Use the book icon portion only, at 32x32px

2. Design Principles

1. Student First — Every decision serves a 17-year-old making a life choice

Use plain, jargon-free language. "Annual fees" not "tuition quantum"
Show the most important information first: fees, package, eligibility
Never bury the counselor contact. It is always one tap away
Avoid dark patterns — no fake countdown timers, no manipulative CTAs

2. Trust Through Data — Every number must be sourced and dated

Show "Last updated: Month Year" on all fee/package data
Display government approval badges (AICTE, UGC, MCI) prominently
Ratings must show review count alongside the number
Use official college logo and banner images, not stock photos

3. Speed is a Feature — Sub-2 second loads on 4G mobile

Lazy-load all images below the fold
Skeleton loaders on all cards during data fetch
No layout shift after images load (set explicit width/height)
Critical CSS inline in <head>. Non-critical deferred

4. SEO is Design — Every page element serves Google indexing

H1 must contain college name + city for every college page
FAQ sections on every college page (drives featured snippets)
Breadcrumb visible on all inner pages
Structured data JSON-LD injected server-side on every page

5. Mobile First — 73% of Indian students browse on phones

Design at 375px width first. Expand to desktop as enhancement
Touch targets minimum 44x44px
Bottom navigation bar on mobile (not hamburger menu)
Sticky "Get Free Counseling" CTA on mobile at all times

3. Component Library

All components are built as reusable React components. Each component spec below includes dimensions, states, colors, and behavior.

3.1 Navigation Bar

The navbar is sticky, always visible. It anchors the brand and provides quick access to the main sections. On scroll, it adds a subtle border-bottom shadow.

Property	Specification
Height	64px desktop / 56px mobile
Background	#FFFFFF with 0.5px bottom border #E5E7EB on scroll
Max width	1280px, centered
Padding	0 24px desktop / 0 16px mobile
Logo area	Logo image 36px height + "EduMadras" text 18px/700 Navy
Nav links	"Colleges" / "Courses" / "Counselors" / "Compare" / "Loans"
Link style	14px/500 #6B7280, hover color #1B3A5C, active color #1B3A5C + underline
Primary CTA	"Get Free Counseling" — Navy filled button, 14px/600
Mobile	Logo left + Hamburger right. Bottom tab bar replaces nav links.
Active state	Current page link: #1B3A5C color + 2px bottom border #1a7a6e
Search trigger	Search icon in navbar opens full-width search overlay

3.2 Button System

Variant	Spec — When to Use
Primary	bg #1B3A5C / text white / 14px/600 / h-44px / r-10px / px-20px — Main actions: Apply, Submit, Enquire
Secondary	border 1.5px #1B3A5C / text #1B3A5C / transparent bg — Supporting actions: View Details, Compare
Teal Accent	bg #1a7a6e / text white — Positive actions: Download Brochure, Check Eligibility
Ghost	no border / text #6B7280 / hover bg #F3F4F6 — Cancel, Close, Dismiss
Danger	bg #DC2626 / text white — Delete, Remove from compare
Hover state	All filled buttons: 8% darker bg. Outline: filled with brand color.

Disabled	opacity 0.4, cursor not-allowed, no hover effect
Loading	Spinner icon left of text, button disabled, text changes to "Sending..."
Icon + Text	Icon 16px, 8px gap, icon on left of text
Full-width mobile	All primary CTAs stretch to 100% on screens under 480px

3.3 College Card

The most important component. Used on listing pages, search results, and homepage. Communicates trust, ranking, and key stats at a glance.

Property	Specification
Width	Grid: 1/3 desktop, 1/2 tablet, 1/1 mobile. Min 280px.
Height	Auto. Banner 160px, body content variable.
Border radius	14px
Background	#FFFFFF
Shadow	0 1px 3px rgba(0,0,0,0.08)
Hover shadow	0 8px 24px rgba(0,0,0,0.12), translateY(-2px), 200ms ease
Banner image	100% width, 160px height, object-fit: cover. Gradient overlay bottom-to-top.
Stream badge	Positioned top-left on banner. Color per stream (see 1.1).
Rank badge	Positioned top-right on banner. "Rank #1" in White/Navy pill.
College logo	48x48px, rounded-lg, white bg, positioned at banner/body boundary.
College name	18px/600 #111827. Truncate at 2 lines.
Location	13px #6B7280, location pin icon 12px
Stats row	Avg Package | Rating | Students — 13px/500, dividers between
Approval badges	AICTE / UGC / MCI chips — 11px/600, each with distinct light bg
CTA row	"View Details" (ghost) + "Get Counseling" (primary) — side by side
Recommended	Gold border 2px #F59E0B + "Recommended" badge top-right when is_recommended=true
Compare checkbox	Appears on hover bottom-left. "Compare" checkbox with count indicator.

3.4 Stream Filter Pills

Property	Specification

Layout	Horizontal scrollable row. No wrapping on mobile.
Options	"All" / "Engineering" / "Medical" / "Management" / "Law" / "Design"
Active state	bg #1B3A5C / text white / 1.5px border #1B3A5C
Idle state	bg transparent / text #374151 / 1.5px border #D1D5DB
Hover	bg #F3F4F6
Size	13px/500, h-36px, px-16px, border-radius 999px
Icons	Each stream has a 14px icon (laptop, stethoscope, briefcase, gavel, pen-tool)
Behavior	Clicking filters college cards immediately (no page reload, client-side filter)
URL update	URL updates to /colleges?stream=engineering for shareability + SEO

3.5 Search Bar

Property	Specification
Variants	Hero (full-width, large) + Navbar (compact, triggered by icon)
Hero size	max-width 640px, h-56px, rounded-xl, white bg, shadow-md
Input	16px/400 text, placeholder "Search colleges, courses, cities..."
Search button	Navy bg, "Search" text, inside the input on right side
Suggestions	Dropdown: Recent searches + Top matching colleges + Courses
Suggestion item	36px height, college logo 20px, college name 14px/500, stream badge
Keyboard nav	Arrow keys navigate suggestions. Enter selects. Escape closes.
Empty state	"No results for X — Try broader search" with 3 suggested streams
Mobile	Full-screen overlay search on mobile. Keyboard auto-opens.

3.6 Counselor Card

Property	Specification
Layout	Horizontal card: Avatar left, Name + Role center, Call button right
Avatar	52px circle, initials if no photo, bg #E6F1FB, text #1B3A5C
Name	15px/600 #111827
Role	13px/400 #6B7280 (e.g. "Senior Counselor — Engineering")
Online indicator	8px green dot on avatar. Shows "Available Now" text when is_active=true

Call button	"Call Now" — Teal filled, phone icon, 13px/600
WhatsApp button	"WhatsApp" — Green outlined, WhatsApp icon, 13px/500
Hover	Entire card slightly elevated. Both buttons show filled state.

3.7 Alert / Announcement Banner

Type	Colors + Usage
Info (default)	bg #E6F1FB / border-left 3px #2A5298 / icon: info circle blue — General announcements
Warning	bg #FAEEDA / border-left 3px #F59E0B / icon: warning triangle amber — Deadlines
Offer	bg #FAEEDA / border-left 3px #F97316 / icon: tag orange — Discounts, scholarships
Success	bg #E1F5EE / border-left 3px #1a7a6e / icon: checkmark teal — Confirmations
Layout	Icon 16px + Title 14px/600 + Message 13px/400 + Dismiss X button
Position	Below navbar, above page content. Stacks if multiple alerts.
With image	When image_url is set: image right side (120px width), text left
Dismissable	X button top-right. State saved in localStorage per alert ID.
Expiry	Automatically hidden after expires_at timestamp. No flash on load.

3.8 Sticky Notes Badge

Sticky notes from the sticky_notes table appear as colored badges on college cards and detail pages. They communicate special attributes like "Top Placements" or "Scholarship Available".

Property	Specification
Size	11px/700 text, px-10px py-4px, border-radius 4px
Color	Driven by color column in sticky_notes table (e.g. #EF4444, #1a7a6e)
Text color	Driven by text_color column (white or dark depending on bg)
Position	Below college name in card, or in a row on detail page
Max visible	3 badges on card, all badges on detail page
With image	If image_url set: small icon 12px before text inside badge

3.9 Breadcrumb

Property	Specification
Style	12px/400 text, chevron separator (>), 8px gaps
Colors	Past crumbs: #2A5298 (clickable link) / Current: #374151 (not linked)
Pattern	Home > Engineering > IIT Madras > B.Tech Computer Science
Mobile	Collapses to "... > [Current Page]" with back arrow on small screens
Schema	BreadcrumbList JSON-LD injected in page <head> matching visible breadcrumb

3.10 Lead Form Modal

Triggered by "Get Free Counseling" or "Apply Now" buttons. Submits to the leads table. This is the primary conversion point of the entire platform.

Property	Specification
Trigger	"Get Free Counseling" / "Apply Now" / "Enquire Now" buttons
Width	480px desktop / full-screen mobile
Overlay	rgba(0,0,0,0.5) backdrop, click outside to close
Header	College logo + "Get Counseling for [College Name]" / generic if no college
Fields	Name (required) / Phone (required) / Email / City / Target Course (dropdown) / Qualification
Phone field	Indian flag emoji + +91 prefix, 10-digit validation, keypad auto-opens mobile
Submit button	"Get Free Counseling" — Full width, Navy primary, loading state
Trust signals	"500+ students counseled" + "Free, no spam" below button
Success state	Replace form with: checkmark + "We will call you within 2 hours"
Prefill	If user is logged in: prefill from users table data
college_id	Automatically attached if modal opened from a college page

3.11 Compare Bar

Appears fixed at bottom of screen when user adds 1+ colleges to compare. Allows comparing up to 3 colleges before navigating to the Compare page.

Property	Specification
Trigger	"Compare" checkbox on college card hover
Position	Fixed bottom, full width, above mobile nav bar
Height	72px

Background	#1B3A5C (Navy)
Content	College thumbnails with X to remove + count "2 colleges" + "Compare Now" button
Max	3 colleges. After 3: "Remove one to add another" toast
Animation	Slides up from bottom on first add. Jiggles on 3rd add.
Compare button	White text, teal bg, navigates to /compare/[slug1]-vs-[slug2]

4. Page Specifications

4.1 Homepage

Purpose	First impression, lead generation, brand trust. Drive users to college discovery or counselor booking.
URL Pattern	`/`

Section 1 — Alert Bar (conditional)

Shown only when active alerts exist in the alerts table. Dismissable. Stacks up to 2.

Section 2 — Hero Section

Property	Specification
Background	Navy-to-Teal gradient (#1B3A5C to #1a7a6e), 100vw, min-height 560px desktop / 480px mobile
Tagline	"Find Your Perfect College Across India" — 48px/700/white
Subtitle	"Compare 500+ colleges · Free expert counseling · Apply online" — 18px/400/white 80% opacity
Search bar	White card, 640px wide, centered below subtitle, large variant (see 3.5)
Trust stats row	Below search: "500+ Colleges" | "14+ Courses" | "Free Counseling" | "6 Streams" in white
Stream quick links	6 stream pills below stats. Light-outlined style on dark bg.

Section 3 — Active Banners / Promotions

Auto-playing carousel sourced from banners table. Max 3 banners. Each has image (right) + title/subtitle/CTA (left). Dots navigation. Auto-advance every 5 seconds. Pause on hover.

Section 4 — Stream Sections (one per stream)

6 sections: Engineering, Medical, Management, Law, Design, in that order. Each section is controlled by show_medical / show_engineering flags in app_settings table.

Property	Specification
Section header	Stream icon + "Top [Stream] Colleges" H2 + "View All" link right
Cards	3 cards on desktop (1 row of 3), 2 on tablet, 1 scroll on mobile. From colleges table filtered by stream, ordered by rank ASC.

Recommended first	Colleges with is_recommended=true appear first within stream.
SEO link	"View All Engineering Colleges" links to /colleges/engineering

Section 5 — Why EduMadras (Trust Section)

3 or 4 feature tiles in a row. "500+ Colleges" / "Free Expert Counseling" / "Verified Data" / "Trusted by 10,000+ Students". Icons + short copy. Light gray background section.

Section 6 — Meet Our Counselors

Grid of counselor cards from counselors table where is_active=true. Max 4 shown. "View All Counselors" link.

Section 7 — FAQ Section

5 most common questions as accordion. Rendered with FAQPage JSON-LD schema. Drives Google featured snippets.

- What is EduMadras?
- How does free counseling work?
- Which entrance exams can I use to search colleges?
- How is fee information verified?
- Can I compare colleges on EduMadras?

4.2 College Listing Page

Purpose	Core discovery page. SEO pillar. Students browse, filter, and shortlist colleges.
URL Pattern	`/colleges/[stream]?city=&type=&exam=&sort=`

Layout

Two-column layout: 280px filter sidebar (desktop) + main content grid. Single column on mobile with filter drawer.

Filter Sidebar

Filter	Type — Options
Stream	Radio pills — Engineering / Medical / Management / Law / Design / All
City	Multi-select checkbox list — From colleges.city values, alphabetical
College Type	Radio — Government / Private / Deemed / All

Entrance Exam	Multi-select — JEE Advanced / JEE Mains / NEET / CAT / CLAT / BITSAT / GMAT / NID DAT / NIFT
Fee Range	Range slider — ₹0 to ₹40L, step ₹1L. Format: "₹5L - ₹20L"
Rating	Star filter — 4.5+ / 4.0+ / 3.5+ / Any
Approvals	Multi-select checkboxes — AICTE / UGC / MCI / NAAC / NBA
Apply button	"Apply Filters" Navy primary + "Reset" ghost — sticky at bottom of sidebar

Main Content

Property	Specification
Results count	"Showing 12 of 47 Engineering Colleges" — 14px/500 #374151
Sort options	Dropdown: "Rank (Low to High)" / "Fees (Low to High)" / "Avg Package (High to Low)" / "Rating (High to Low)"
College cards	3-column grid desktop, 2-col tablet, 1-col mobile. Card component from 3.3.
Pagination	12 per page. Numbered pagination: Prev / 1 2 3 ... / Next. URL updates: ?page=2
Empty state	If no results: Illustration + "No colleges match your filters" + "Reset Filters" button
SEO H1	"Top Engineering Colleges in India 2025" — visible H1 above results count
SEO intro	2-3 sentences about the stream, written per-page. Below H1, above filters.

4.3 College Detail Page

Purpose	The most content-rich, SEO-critical page. Every data point from the schema becomes indexed content.
URL Pattern	`/colleges/[stream]/[slug]`

Above the fold — College Hero

Property	Specification
Banner	Full-width 320px height. banner_url from colleges table. Gradient overlay.
Logo	80px, white card bg, positioned bottom-left of banner overlapping body
College name	H1, 32px/700 #111827. Must include city: "IIT Madras, Chennai"

Breadcrumb	Home > Engineering > IIT Madras
Stats bar	Rank #1 | Rating 4.9/5 | Est. 1959 | 10,000 Students — horizontal, dividers
Approval chips	AICTE · UGC · NAAC A++ — colored chips
Sticky notes	Colored badges from college_sticky_notes. E.g. "Top Placement 2024"
CTA buttons	"Get Free Counseling" (primary) + "Download Brochure" (secondary) + "Add to Compare" (ghost)

Page Layout — Sticky Sidebar

Left column: main content (Courses, About, Placements, FAQ). Right column (320px sticky): Lead form + Counselor card + Compare button. On mobile: lead form floats as sticky CTA at bottom.

Main Content Sections (in order)

Section	Content Details
Courses & Fees	Table: Course Name | Duration | Seats | Annual Fee | Eligibility | Avg Package. Sourced from college_courses JOIN master_courses. Rows expand for details.
About the College	description from colleges table. 2-3 paragraphs. Followed by key facts: Type, City, State, Established Year.
Placement Statistics	avg_package and highest_package from colleges table. Bar chart or stat cards. "Top Recruiters" placeholder section.
Admission Process	Eligibility criteria from college_courses table, formatted per course. Step-by-step numbered list.
Campus & Facilities	Text + image gallery. Editorial content section (can be CMS-driven later).
FAQ Section	5+ Q&A pairs specific to this college. Accordion. FAQPage JSON-LD schema.
Similar Colleges	3 college cards from same stream. "Students also viewed" heading.

4.4 Compare Page

Purpose	Side-by-side comparison of 2-3 colleges. High-intent traffic. Programmatically generated from URL params.
URL Pattern	`/compare/[slug-a]-vs-[slug-b]`

Section	Spec

Header	"IIT Madras vs IIT Delhi — Which is Better?" — H1, includes both college names for SEO
College columns	2-3 columns, one per college. College banner + logo + name at top of each.
Comparison table	Rows: Rank / Rating / Fees / Avg Package / Highest Package / Total Students / Type / Approvals / Location / Established. Color-code winner per row (green bg).
Courses comparison	Common courses shown side by side with fee differences highlighted.
Winner indicator	Small trophy icon next to better value in each row. Configurable logic.
Editorial verdict	"Which is better for Computer Science?" — Short editorial paragraph below table.
CTA	"Get counseling to choose between these colleges" — Navy banner section below table.
Schema	Both colleges as EducationalOrganization in JSON-LD. BreadcrumbList.

4.5 Course Detail Page

Purpose	All colleges offering a specific course. High SEO value. Sourced from master_courses + college_courses.
URL Pattern	`/courses/[course-slug]`

Property	Spec
H1	"B.Tech Computer Science — Colleges, Fees & Eligibility 2025"
Course overview	Duration, typical eligibility, stream. 2-3 sentences.
Colleges table	All colleges offering this course. Columns: College | City | Type | Fee | Eligibility | Avg Package | Action.
Sort	By fee / by rank / by package. Default: rank ASC.
Filter	Government/Private filter. City filter.
Eligibility info box	Which entrance exams qualify. Sourced from eligibility column.

4.6 Entrance Exam Page

Purpose	All colleges accepting a specific entrance exam. Massive seasonal traffic (Jan-June).
URL Pattern	`/entrance-exams/[exam-slug]`

Property	Spec
H1	"Colleges Accepting JEE Advanced 2025 — Fees, Cutoffs & Counseling"
Exam info box	What is this exam, who conducts it, when results come. 100-150 words.
College grid	All colleges with this exam in eligibility column. Full card grid with stream filter.
Cutoff section	Placeholder table for cutoff ranks by category (General/OBC/SC/ST). Add-on feature.
Important dates	Timeline component: Registration / Exam / Result / Counseling dates.

4.7 City Landing Page

Purpose	All colleges in a specific city. Local SEO powerhouse.
URL Pattern	`/cities/[city-slug]`

Property	Spec
H1	"Top Colleges in Chennai 2025 — Engineering, Medical & More"
Intro text	Brief description of the city's education ecosystem. 100 words.
Stream tabs	Filter colleges by stream. All / Engineering / Medical / Management etc.
College grid	Full card grid. Filtered from colleges.city column.
City stats box	Number of colleges | Streams available | Govt vs Private ratio.

4.8 Counselors Page

Purpose	Browse all counselors and initiate contact. Trust-building page.
URL Pattern	`/counselors`

Property	Spec
H1	"Talk to Our Expert College Counselors — Free Guidance"
Intro	2 sentences: "Our counselors are former admissions officers and education experts."
Grid	3-col desktop / 2-col tablet / 1-col mobile. Counselor cards from counselors table.
Filter	Stream specialization filter: Engineering / Medical / Management / All.
CTA top	Wide banner: "Not sure which counselor to choose? We'll match you." + form.

Availability badge	Green "Available Now" or Gray "Busy" based on is_active field.

5. Mobile & Responsive Design

73% of Indian students browse education sites on mobile. Mobile is the primary design target, not a secondary concern.

Breakpoint	Width — Layout Changes
Mobile S	320px — Single column. Simplified nav. Compact cards.
Mobile L	375px — Primary mobile design target. All mobile specs apply here.
Mobile XL	428px — Same as Mobile L, minor spacing increases.
Tablet	768px — 2-column cards. Sidebar as drawer. Tablet nav.
Desktop S	1024px — 3-column cards. Sidebar visible. Full nav.
Desktop L	1280px — Max-width container. Content centered.
Desktop XL	1440px+ — No change. 1280px max-width, outer margin fills.

Mobile-specific UI Patterns

- Bottom navigation bar: 5 tabs — Home / Colleges / Compare / Counselors / Profile. 60px height, white bg, border-top.
- Sticky CTA bar: "Get Free Counseling" fixed bottom, above nav bar, 100% width, Navy. Only on college detail pages.
- Filter drawer: Slides up from bottom 85% screen height. Scrim behind. Drag handle at top.
- Search: Full-screen overlay. Auto-focuses input. Large touch targets (44px min).
- College cards on mobile: Full width. Horizontal scroll layout for stream sections.
- Compare bar: Above bottom nav when active. Compact 3-thumbnail layout.
- Lead form: Full screen modal. Keyboard-aware (viewport adjusts for keyboard).

6. SEO UI Patterns

These are UI elements that directly impact Google search ranking and appearance. Every element here is non-negotiable.

6.1 Page Title Formula (in <title> tag)

```
[College Name] — Fees, Courses, Placements [Year] | EduMadras
Top Engineering Colleges in India [Year] — Rankings, Fees | EduMadras
Colleges Accepting JEE Advanced [Year] — List, Cutoffs | EduMadras
```

6.2 Meta Description Formula

```
IIT Madras: B.Tech fees ₹8L/yr, avg package ₹21.5L. Rank #1 Engineering.
AICTE approved. Compare courses, check eligibility, get free counseling.
```

6.3 H1 Placement Rule

Every page must have exactly one H1. It must appear in the visible page content above the fold. It must contain the primary keyword for that page.

6.4 FAQ Component (Required on every college page)

5+ question-answer pairs. Rendered as visible accordion on page. FAQPage JSON-LD injected in <head>. Questions should be exact phrases students search.

- What is the [course] fee at [College]?
- What is the cutoff for [College]?
- Is [College] government or private?
- What is the placement package at [College]?
- Which entrance exam is required for [College]?

6.5 Breadcrumb (Required on all inner pages)

Visible breadcrumb component as described in 3.9. BreadcrumbList JSON-LD must exactly match what is visible on screen.

6.6 Last Updated Timestamp

"Data last verified: March 2025" must appear on every college detail page, near the fees table. This is an E-E-A-T signal Google values.

6.7 Internal Linking Requirements

- Every stream badge on a college card must link to /colleges/[stream]
- Every city mention must link to /cities/[city-slug]
- Every entrance exam mention must link to /entrance-exams/[exam-slug]
- College detail page must link to 3 "Similar Colleges" cards
- Compare page must link back to both college detail pages

7. Interaction States & Micro-animations

Element	Interaction — Animation Spec
College card hover	translateY(-4px) + shadow deepens. 200ms ease-out. Cursor: pointer.
Button hover	Background 8% darker. 150ms ease. Scale stays 1.
Button active/click	scale(0.97). 100ms. Releases to 1 on mouse-up.
Button loading	Spinner replaces icon. Text stays. No layout shift. Disabled state.
Filter pill select	Background fills Navy. Text turns white. 150ms ease.
Stream section scroll	Horizontal scroll with momentum. Hide scrollbar but keep functionality.
Modal open	Backdrop fades in 200ms. Modal slides up 12px + fades in 250ms.
Modal close	Backdrop fades out 150ms. Modal fades down 200ms.
Toast notification	Slides in from bottom-right. Auto-dismiss after 4s. Manual X dismiss.
Compare bar appear	Fixed bar slides up from bottom 300ms ease-out on first item add.
Page transitions	Next.js default. Fade: 150ms on exit, 200ms on enter.
Skeleton loaders	All card grids show skeleton cards while fetching. Pulse animation 1.5s.
Search suggestions	Dropdown fades in 150ms. List items stagger 30ms each.
Accordion FAQ	Content height animates open/close. 250ms ease. Chevron rotates 180deg.
Image lazy load	Blur placeholder (blurDataURL from next/image). Fades in when loaded.

8. Accessibility Requirements

EduMadras must meet WCAG 2.1 AA standard. Students with disabilities deserve equal access to education opportunities.

Requirement	Implementation
Color contrast	All text on colored backgrounds must meet 4.5:1 ratio minimum. Navy on white: passes. White on Navy: passes. Check all badge text.
Focus indicators	All interactive elements: 3px solid #1a7a6e outline offset 2px on keyboard focus. Never remove outline with outline:none only.
Alt text	All college logo images: alt="[College Name] Logo". Banners: alt="[College Name] campus". Decorative: alt=""
Semantic HTML	Navigation: <nav>. Main content: <main>. College sections: <section> with aria-label. Sidebar: <aside>. Headings in correct H1>H2>H3 order.
Touch targets	Minimum 44x44px for all interactive elements on mobile. Apply min-h-[44px] to all buttons.
Form labels	All inputs have visible <label> or aria-label. Required fields have aria-required="true". Errors use aria-describedby.
Keyboard navigation	Tab order follows visual reading order. Modal traps focus inside. Escape closes modal/drawer.
Screen reader	Stream badges: aria-label="Engineering college" not just the color. Stats: aria-label="Average placement package: 21.5 lakh rupees".
Loading states	aria-busy="true" on containers during data fetch. aria-live="polite" on result counts when filters change.
Skip link	"Skip to main content" as first focusable element. Visible on focus.